



國立政治大學資訊管理學系

NCCU DEPARTMENT OF MANAGEMENT INFORMATION SYSTEMS

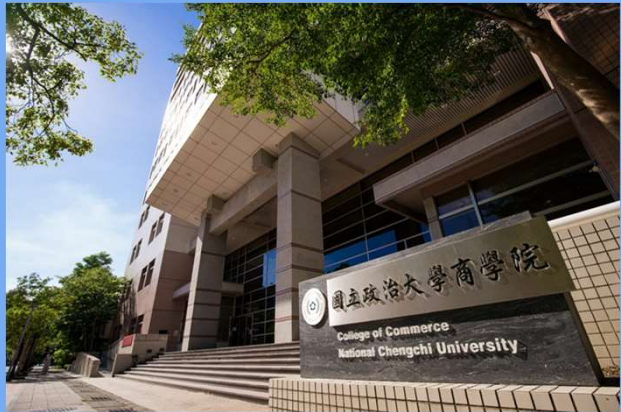
Introduction to Computer Science

Midterm Review

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MIDTERM

Date: Oct. 31 (Fri)

Time: 9am~12pm (306005001)

Time: 1pm~4pm (306005011)

Location: 逸仙5樓資管電腦室

**Before it is too late,
let's do a quick review...**

Types of Questions 1

True/False: mark T for True and F for False.

If False, rewrite the statement so that it is True. (either Chinese or English)

Sample Question:

(**F**) The Internet is a service of the World Wide Web (WWW).

If false: **WWW is a service of Internet**

Types of Questions 2

Multiple Choice: only ONE correct answer

(單選題: 選項很多, 真相只有一個)

Sample Question:

The Internet is the world's largest _____.

- a. LAN
- b. WAN
- c. FAN
- d. PAN

Types of Questions 3

Matching: match the terms with their definitions

Sample Question:

(a) thrashing (b) queuing

__b__ Process manager uses waiting lists to store information

__a__ OS spends too much time on paging

Types of Questions 4

Open Questions:

answer the questions in either Chinese or English

Sample Question:

Explain the difference between DSL and Cable services

Notes



Midterm is a good time to examine your learning performance. The point is to figure out whether you understand the concepts/components rather than asking you to memorize all the definitions.

Therefore, you should know what this concept/term/component is about and be able to select the best answer from the question itself (true/false, multiple choice, and matching). For the Open question, briefly summarizing your answers would work. The key is to point out the differences/pros/cons and provide appropriate answers to show us that you fully understand the concepts/logic of it.



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Introduction to Computer Science

Computer Hardware

Shih-Yi (James) Chien

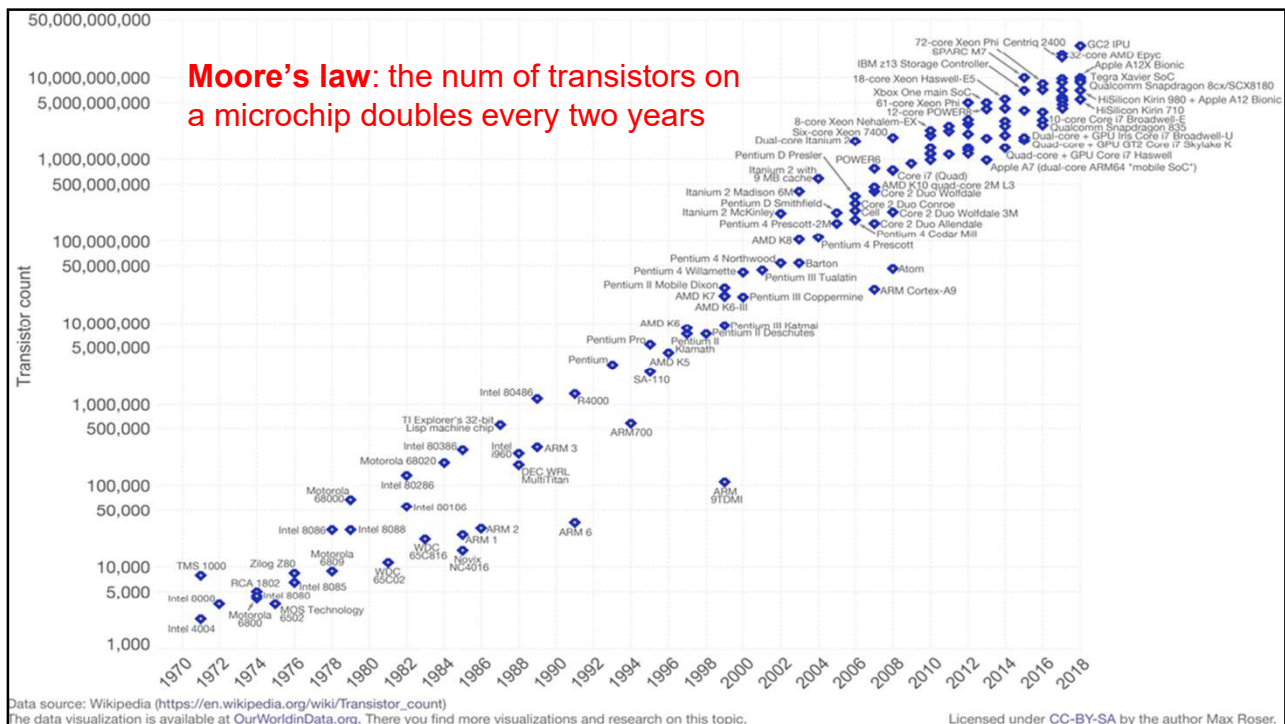
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Key Points

1. What is Moore's law
2. What is ALU and CU? What are the differences?
3. What is Machine cycle? How does it work?
4. Why do we need Cache and Register? What are the differences?
5. **Computer Memory Hierarchy**
6. What are the differences b/t a Hard Disk and SSD
7. Storage Capacity



CPU Components- ALU & CU

CPU performs operations on data, which has three parts: an **arithmetic logic unit (ALU)**, a **control unit (CU)**, and a set of **registers**

When running an application:

Instructions data are transferred from storage device to memory

CU directs and coordinates the instruction flow in the computer

- CU interprets and executes the instructions in the memory
- CU controls ALU, access to memory and I/O systems

ALU performs logic, shift and arithmetic operations

- ALU performs calculations on data in memory (logic, shift, and arithmetic operation)

Data are stored in **register** for faster access

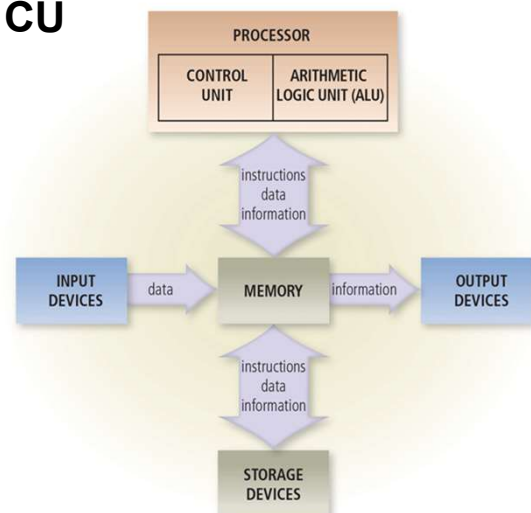


Figure 6-4 Most devices connected to the computer communicate with the processor to carry out a task.



CPU Components- Register

Registers are fast stand-alone storage locations that hold data temporarily

- Multiple registers to facilitate CPU operation

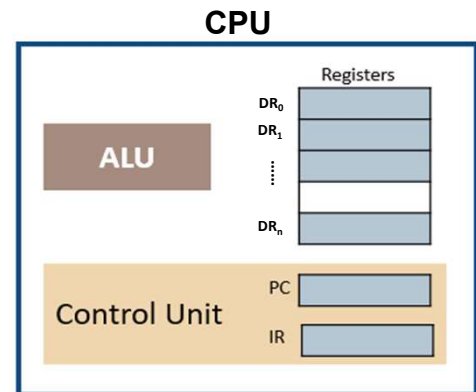
DR (data register): hold the input data, the intermediate results, and the final result of operations

- Numerous DR are inside the CPU to speed up the operations

IR (instruction register): CPU fetches instructions one by one from memory and stores them in IR

PC (program counter): keep track of the instruction currently being executed and to be executed next

- After execution of the instruction, the counter is incremented to point to the address of the memory location of the main memory that holds the next program instruction



Different registers

DR: data register

IR: instruction register

PC: program counter



Machine Cycle

Machine cycle is the activity performed by CPU to execute a program instruction

For each instruction, the processor repeats a set of four basic operations

Fetch: obtain data or an instruction from memory

- CU orders the system to copy next instruction into IR in CPU
- The address of the instruction to be copied is held in PC register
- After copying, PC is incremented to refer to the next instruction in memory

Decode: translate instructions into signals the computer can execute

- CU decodes the instruction stored in IR and send data and instructions to ALU

Execute: carry out the commands

- CU can tell the system to load a data item from memory, or
- CU can tell ALU to perform calculation (eg, add two inputs)

Storage: store the results in memory

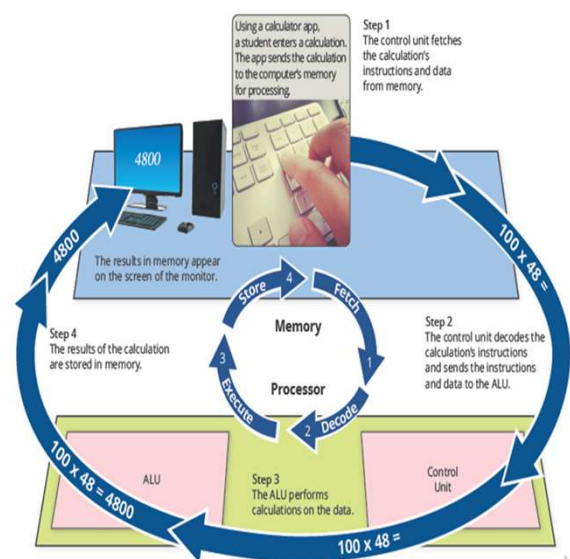


Image credit: Discovering Computers 2018: Digital Technology, Data, and Devices



Example of Machine Cycle

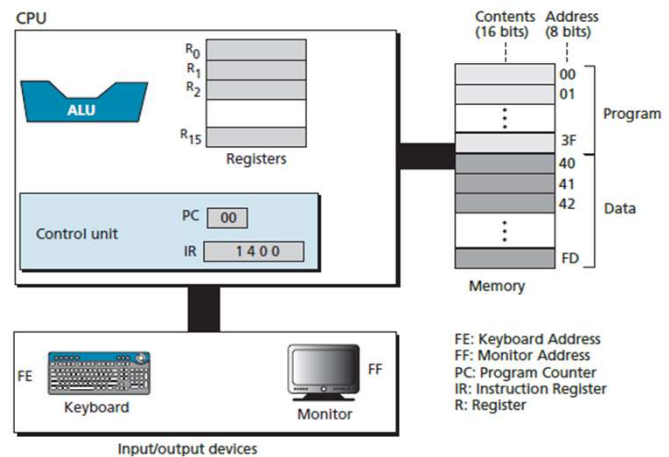
Example: $161 + 254 = 415$

These numbers are shown in the hexadecimal format in memory:

$161 = (00A1)_{16}$

$254 = (00FE)_{16}$

$415 = (019F)_{16}$

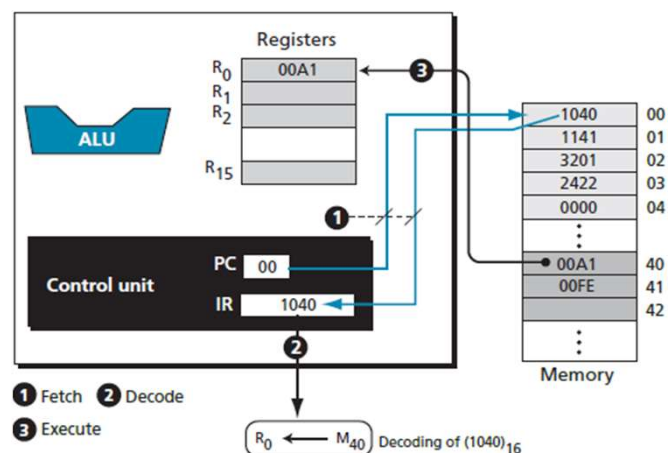


Cycle 1

The PC points to the first program instruction, memory location $(00)_{16}$

The CU goes through 3 steps:

1. CU **fetches** the instruction stored in memory location $(00)_{16}$ and puts it in the IR. Then, PC is incremented $(00 \rightarrow 01)$
2. The CU **decodes** the instruction $(1040)_{16}$ as $R_0 \leftarrow M_{40}$
3. The CU **executes** the instruction, a copy of the integer stored in memory location (40) is loaded into R_0 , $(00A1)_{16}$



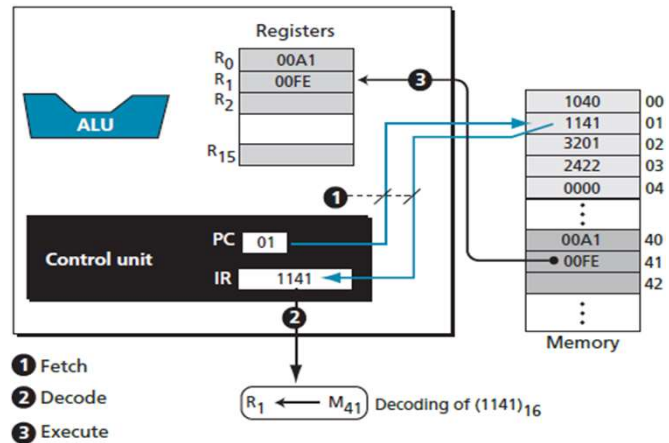


Cycle 2

The PC points to the first program instruction, memory location $(01)_{16}$

The CU goes through 3 steps:

1. CU **fetches** the instruction stored in memory location $(01)_{16}$ and puts it in the IR. Then, PC is incremented $(01 \rightarrow 02)$
2. The CU **decodes** the instruction $(1141)_{16}$ as $R_1 \leftarrow M_{41}$
3. The CU **executes** the instruction, a copy of the integer stored in memory location $(41)_{16}$ is loaded into R_1 , $(00FE)_{16}$



Cache Memory

Speed: Register > Cache > Main memory

Cache is placed between CPU and main memory

Cache stores frequently used instructions and data (a copy of a portion of main memory)

- Data might be needed again soon
- Speed up processing times

When CPU needs to access a word in main memory:

1. CPU checks cache
2. If the word is there, CPU copies the word.
If not, CPU accesses main memory and copies a block of memory starting with the desired word.
This block replaces the previous contents of cache memory
3. CPU access the cache and copies the word

L1 & L2 cache built directly on the processor chip

L3 cache is on the motherboard

- L1: fastest but smallest; L3: largest but slowest

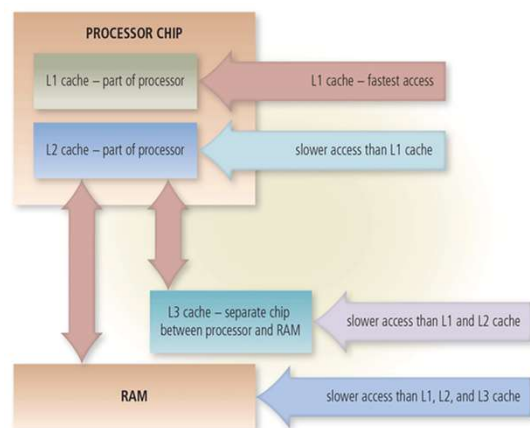


Figure 6-14 Memory cache helps speed processing times when the processor requests data, instructions, or information.



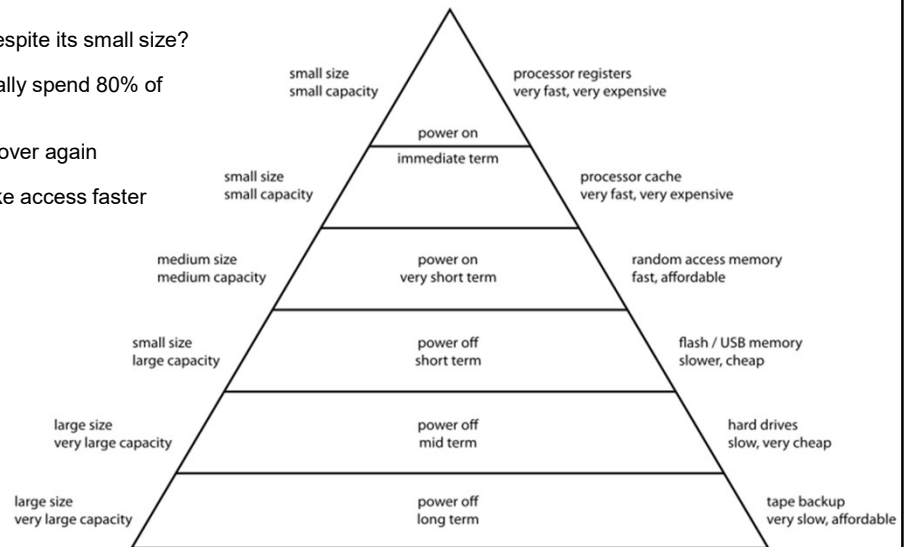
Computer Memory Hierarchy

Why (cache) memory is so efficient despite its small size?

The **80:20** rule: most computers typically spend 80% of time accessing 20% of the data

The same data is accessed over and over again

Cache memory holds this 20% to make access faster



Characteristics of Hard Disk

Before reading from or writing on a hard disk, the disk must be formatted

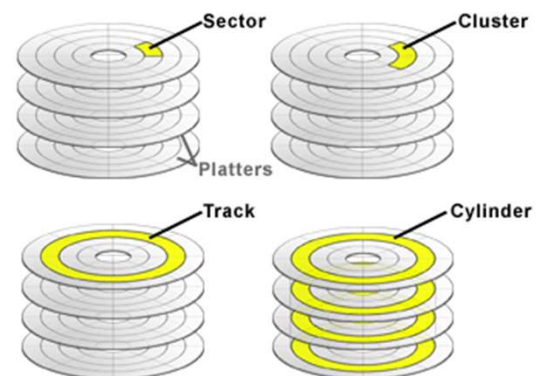
Track is a narrow recording band that forms a full circle on the surface of a disk

- **Cylinder**: tracks that line up on each platter from top to bottom and can be read at the same time

Breaking the tracks into small arcs called **sector**

- A sector stores up to 512 bytes of data
- Several sectors ($n > 1$) form a **cluster**

Revolutions per minute (RPM) : the number of turns in one minute





Solid State Drive (SSD)

SSD (solid state drive) is a flash memory storage device that contains its own processor to manage its storage

An SSD has several advantages over traditional (magnetic) hard disks:

- Faster access times
- Faster transfer rates
- Quieter operation
- More durable
- Lighter weight
- Less power consumption
- Less heat generation
- Longer life
- Defragmentation not required

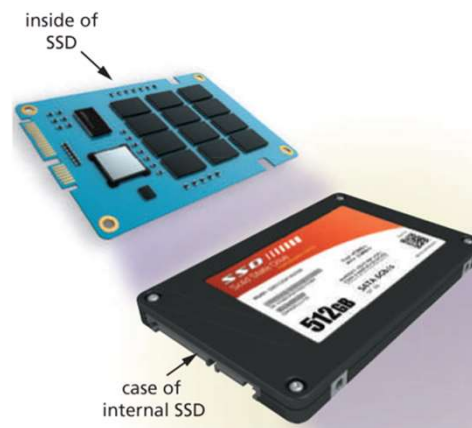


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Storage Capacity

Storage Term	Approximate Number of Bytes	Exact Number of Bytes
<i>Kilobyte (KB)</i>	1 thousand	2^{10} or 1,024
<i>Megabyte (MB)</i>	1 million	2^{20} or 1,048,576
<i>Gigabyte (GB)</i>	1 billion 1GB=1000MB	2^{30} or 1,073,741,824
<i>Terabyte (TB)</i>	1 trillion 1TB=1000GB	2^{40} or 1,099,511,627,776
<i>Petabyte (PB)</i>	1 quadrillion	2^{50} or 1,125,899,906,842,624
<i>Exabyte (EB)</i>	1 quintillion	2^{60} or 1,152,921,504,606,846,976
<i>Zettabyte (ZB)</i>	1 sextillion	2^{70} or 1,180,591,620,717,411,303,424
<i>Yottabyte (YB)</i>	1 septillion	2^{80} or 1,208,925,819,614,629,174,706,176

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Introduction to Computer Science

Operating system

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Key Points

1. Roles of different OS components
2. Differences b/t Monoprogramming and Multiprogramming
3. The uniqueness among different types of memory management in Multiprogramming
4. Definition of different components in process manager
5. How/when does a status change in the state diagram?
6. What is a Deadlock/Starvation in OS? How does it happen?



Components of OS

Shell is a user interface for access to an OS's services

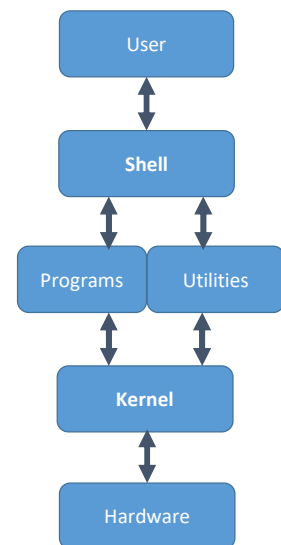
- User give shell a command
- Command-line interface or graphical user interface
- OS can have several different shells

Utilities provide a support process for users

- Common utilities: text editors, search programs, sort programs, etc.

Kernel is the heart of an OS with complete control over everything in the system

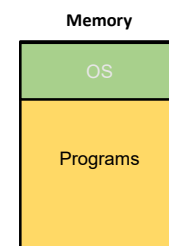
- Contain the most basic parts of OS
- Memory management, process management, device/file management
- Other components of the system call on the kernel to perform services
- If the command requires an application, the shell requests kernel to run it



Memory Management

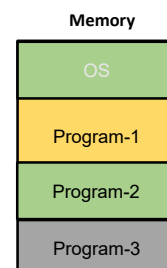
Monoprogramming (belongs to the past)

- Memory capacity is dedicated to a **single program**
 - Only little part is needed to hold the operating system
- When one program is running, **no other program be executed**
 - Speed: CPU >> Input & Output
 - CPU is idle when receiving data from or sending data to devices
- If size of program > size of memory: cannot be run



Multiprogramming (current approach)

- Multiple programs are stored in memory and executed concurrently
 - **CPU switch** rapidly between programs

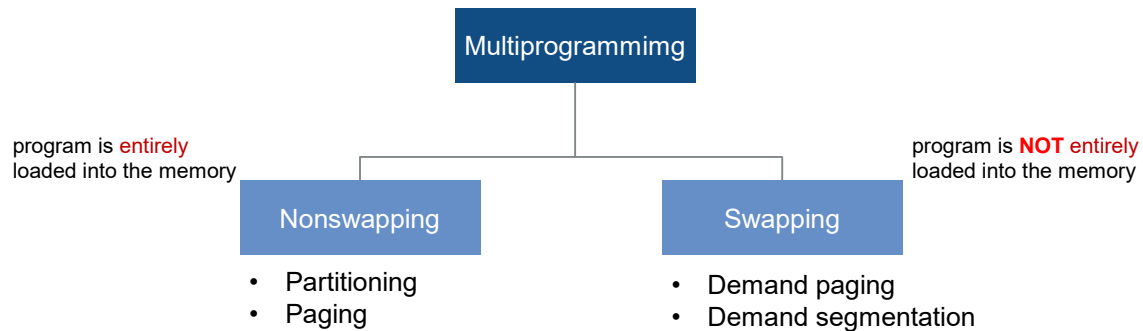




Memory Management- Multiprogramming

Nonswapping: program keeps in memory for the duration of execution

Swapping: programs can be swapped between memory and disk



Nonswapping- Partitioning

Memory is divided into **variable-length** sections

- Each partition holds one program
- CPU switches between programs
 - Execute instructions of the program, until an I/O operation is encountered or the time allocated for the program expires
- Each program is **entirely loaded** into the memory, requiring **contiguous** locations
 - Small partition size: programs cannot be loaded into memory
 - Large partition size: holes (unused locations) in memory
 - Memory manager can compact the partitions to remove holes and create new partitions, but it takes extra costs

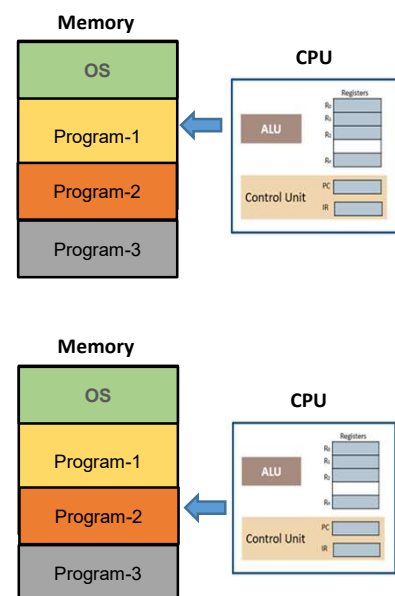


Image credit: Foundations Of Computer Science 4th edition, Behrouz Forouzan, (ISBN: 1473751047)



Nonswapping- Paging

Programs are divided into equally sized sections:
pages

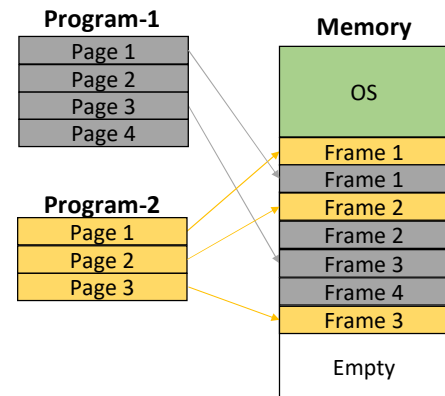
Memory is divided into equally sized sections: **frames**

- The size of a page/frame is the same and equal to the size of the block used by the system

Programs do not have to be contiguous in memory

- Two consecutive pages can occupy **noncontiguous** frames in memory

Paging can improve efficiency, but the **entire** program **needs to be loaded** into memory before execution

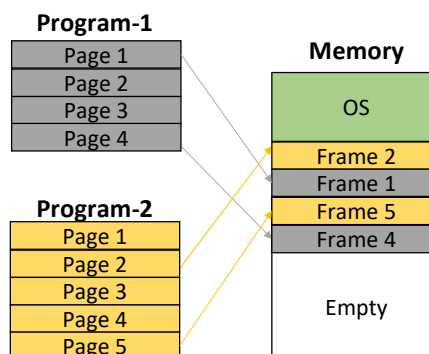


Swapping- Demand Paging

A program is divided into pages, and these pages can be loaded into memory **one by one (not entirely)**, and can be executed and replaced by another page

Memory can hold pages from multiple programs at the same time

- Pages can be loaded into any free frame

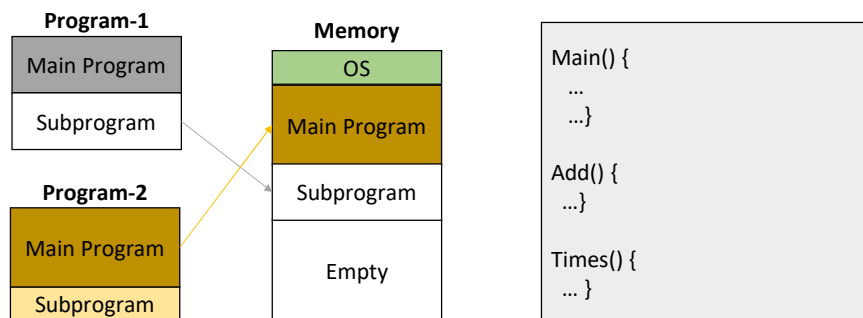




Swapping- Demand Segmentation

A program is usually made up of a **main program and subprograms**

A program is divided into **multiple segments**, and the segments are loaded into memory, executed and replaced by another module in the same or different program



Process Manager

Program: a nonactive set of instructions stored on disk

- A program may or may not become a job

Job: a program becomes a job when it is selected for execution

- When finished executing, a job becomes a program again
- Every job is a program, but not every program is a job

Process: a program in execution (has started but has not finished)

- A job is being run in memory
- Selected among other waiting jobs and loaded into memory
- Every process is a job, but not every job is a process
- One process can have multiple threads

Process Name	% CPU	CPU Time	Threads
Firefox	22.8	11:27.14	85
Spotify	2.2	11.71	42

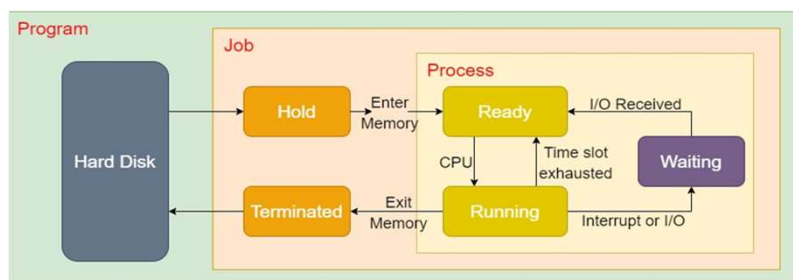


State Diagram

1. Program becomes Job when selected by OS and bring to Hold state
2. Once being loaded to **memory**, the Job moves to **Ready** state and **becomes Process**
3. When the CPU can **execute** the Job, it moves to **Running** state

In **Running** state, three things can happen:

- Process execution until I/O are needed → move to **Waiting** state until I/O is complete
- Process exhausts its allocated time slot → move to **Ready** state
- Process terminates → move to **Terminated** state



Queuing for Scheduling

Process manager uses queues (**waiting lists**) to store information

- Job/Process control block (PCB): store information about jobs or processes
- Process manager stores the job/process control block
- Job or process remains in memory (can be too large to be saved in a queue)

An OS can have several queues

- Job queue: hold jobs that are waiting for memory
- Ready queue: hold processes that are in memory, ready to be run, and waiting for CPU

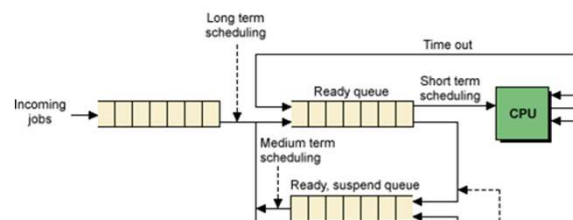


Image credit: <https://www.technologyuk.net/computing/computer-software/operating-systems/process-management.shtml>



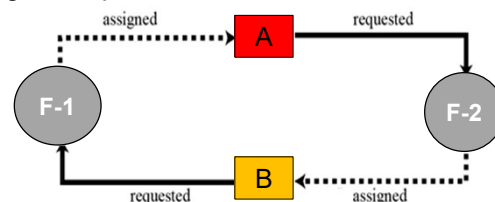
Deadlock

Deadlock occurs when the OS **fails to put resource restrictions** on processes

- If the OS allows the process to start running without first checking whether the required resources are ready
 - To avoid: cannot start running until the required resources are free
- If the OS allows the process to **reserve resources** as needed **without restrictions**
 - To avoid: limit the time a process can hold a resource

When resources are accessed by multiple users

- File-1 is assigned to process-A and cannot release until it acquires File-2
- File-2 is assigned to process-B and cannot release until it acquires File-1



Starvation

Starvation is the opposite of deadlock

- When OS **puts too many resource restrictions** on a process

Process-A needs file-1 & file-2

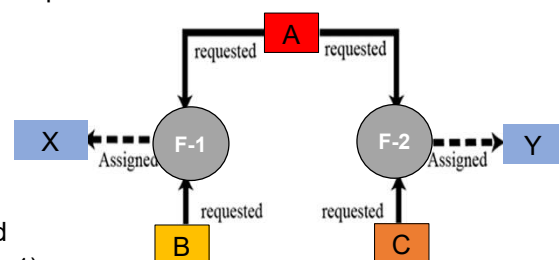
- File-1 is being used by process-X
- File-2 is being used by process-Y

Process-X terminates and release File-1

- Process-A cannot start as File-2 is still occupied
- Process-B is allowed to run (only requested File-1)

Process-Y terminates and release File-2

- Process-A cannot start as File-1 is still occupied
- Process-C is allowed to run (only requested File-2)





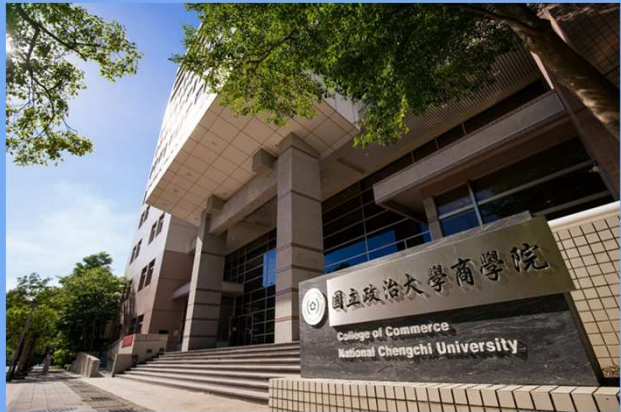
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Introduction to Computer Science

Network

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Key Points

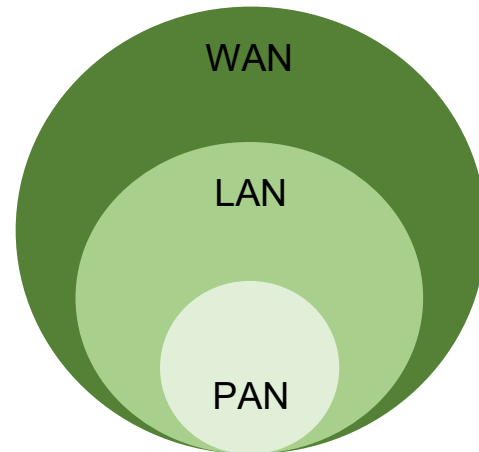
1. Differences among WAN vs. LAN vs. PAN
2. Differences among Intranet vs. Internet vs. Extranet
3. Differences and Pros/Cons b/t Client-Server and P2P architecture
4. Differences and Pros/Cons among different network topology
5. What is Mac/IP address? What are the differences/uniqueness?
6. How does internet work? (say google a keyword via a browser)
7. Differences among Hub vs. Switch vs. Router
8. Differences among types of RAID



What is a Network

Computer network to **connect to each other**

- Wide area network (WAN)
- Local area network (LAN)
- Personal area network (PAN)



Personal Area Network (PAN)

PAN is a network for interconnecting devices centered on an **individual's** workspace

- A network that connects computers and devices in an **individual's workspace** through wired and wireless technologies
- Devices on PAN are usually connected via Bluetooth
 - Bluetooth : a short-range wireless technology
 - Physical range: typically less than 10 m



Local Area Network (LAN)

LAN is a network that connects computers and devices in a **limited geographical area**

- Home, school, office building, departments, etc.

Each computer or device on the network, called a **node**, which often shares resources

- Printers, hard drives, etc.

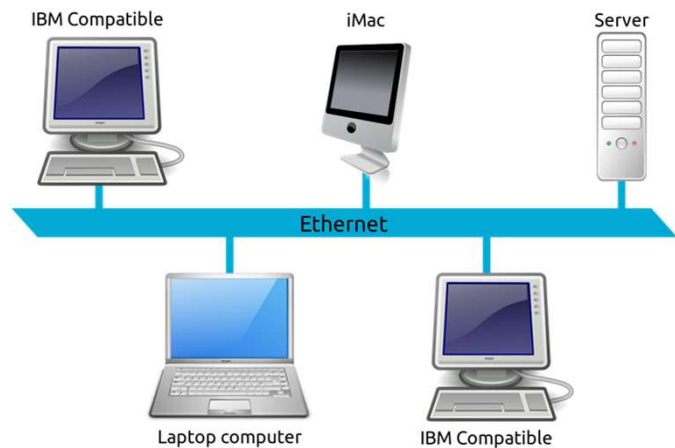


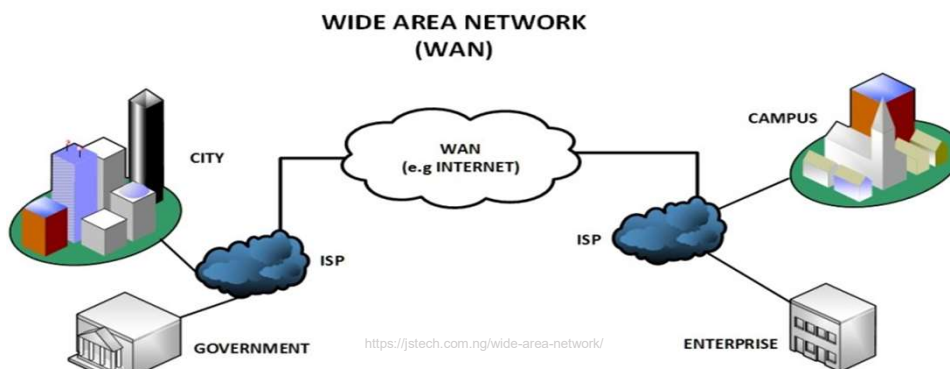
Image credit: https://en.wikipedia.org/wiki/Local_area_network#/media/File:Ethernet_LAN.svg



Wide Area Network (WAN)

Wide Area Network (WAN) is a telecommunication network that extends over a large geographic area for the primary purpose of computer networking

- The Internet may be considered a WAN



<https://jstech.com.ng/wide-area-network/>



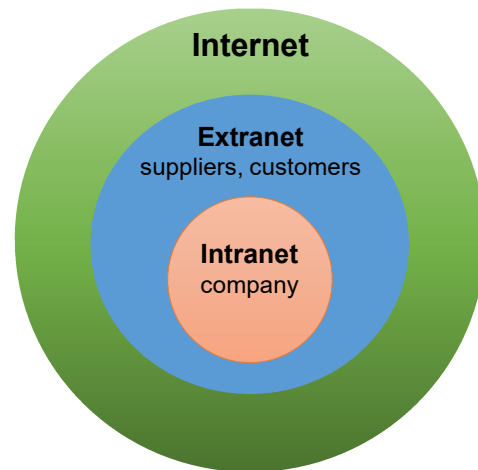
Computer Network for Sharing Information

Intranet

- A private network for authorized individuals
- Companies use the intranet to communicate internally

Extranet

- Allows outsiders (such as customers and suppliers) to access an organization's intranet
- A supplier can check the customer's inventory level before deciding whether to ship other products



	Internet	Extranet	Intranet
Type of Network	Public	Private	Private
Accessibility	Anyone	<u>Authorized</u> people	<u>Authorized</u> people
Size	<u>Large</u> number of connected to the devices	Limited number of connected devices over internet	Limited number of connected devices
Information Sharing	Information can be shared across the world	Information can be shared b/t employees and external people	Information can be shared securely within an organization
Example	World Wide Web, Social media, Email	Network of collaboration b/t corporations	Internal operations within an company



Network Architecture- Client/Server

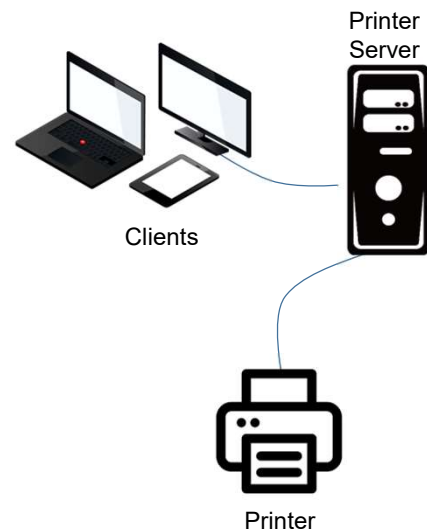
Server: provide service(s)

- Centralized storage location which is accessible to other computers on the network

Client: other computers on the network request resource from the server(s)

- Rely on the server for its resources
- Different clients may have different permissions to access files or resources

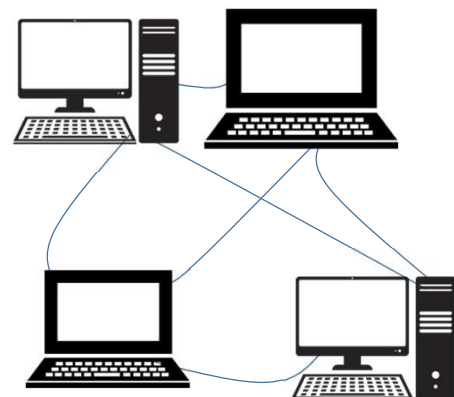
Can connect to one or more servers to sharing files or resources



Network Architecture- P2P

Peer-to-peer (P2P) network

- Computers **communicate directly with each other** and share each other's resources
 - Computer-A uses a printer connected to Computer-B and revising a file stored on Computer-C
- Administrator is not required since the P2P network treats all computer equally



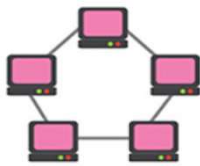


Network Topology



Bus topology

All devices attach to a central cable (called bus) to transfer data
If the **bus** fails, the devices on the network will not be able to communicate



Ring topology

Data is transferred sequentially from one device to another
If **one of the devices** on the network fails, the communication is no longer available



Network Topology



Star topology

- Every device on the network is connected to the central device (server/switch/hub)
- If the central device fails, other devices will not be able to communicate
- The bus can also be used to connect multiple star networks together to form a tree topology
- Tree topology is usually used in schools and enterprises



Mesh topology

- All devices interconnect with each other
- If a single device on the network fails, the rest of the network will communicate through the alternate route to continue operation
- Full mesh topology: each device on the network is connected to all other devices on the network



IP address and Domain Name System

An IP address is a sequence of numbers that uniquely **identifies each computer** or device's location to connected to the Internet or any other network

The **domain name** is a text-based name which **corresponds to the IP address** of the server

The **Domain Name System (DNS)** server **converts** the domain name to its associated IP address

IPv4 address: 74.125.22.139

IPv6 address:
2001:4860:4860::8844

Domain name: google.com

	IPv4	IPv6
Full Name	Internet Protocol version 4	Internet Protocol version 6
Format	32-bit Internet addresses	128-bit Internet addresses
Capacity	2^{32} IP addresses (4.29 billion)	2^{128} IP addresses



How does internet work?

1. Input google.com in the browser's address bar
2. Browser asks DNS server: how to reach google.com (DNS server is maintained by service providers)
3. DNS looks up domain name and its associated IP address
4. DNS replies: go to 172.217.27.142
5. Browser uses the IP address and contact the Google web server to request the content
6. Google server checks with its DB regarding the requested content
7. DB finds the relevant information and responds to Google server
8. Google server sends the requested content to the user's browser (webpages)
9. Browser shows whatever it receives from the Google server on the screen

```
C:\Users\admin>nslookup google.com
伺服器: sun2cc.nccu.edu.tw
Address: 140.119.1.110

未經授權的回應:
名稱: google.com
Addresses: 2404:6800:4008:803::200e
172.217.27.142
```



https://www.youtube.com/watch?app=desktop&v=7_LPdtIKXPc



Elements and Devices to Create a Network

Modem: the communications device that connects a communications channel to a device

Hub: a central point in a network; transmit data to all devices

Switch: a central point in a network; only transmits data to the intended device(s)

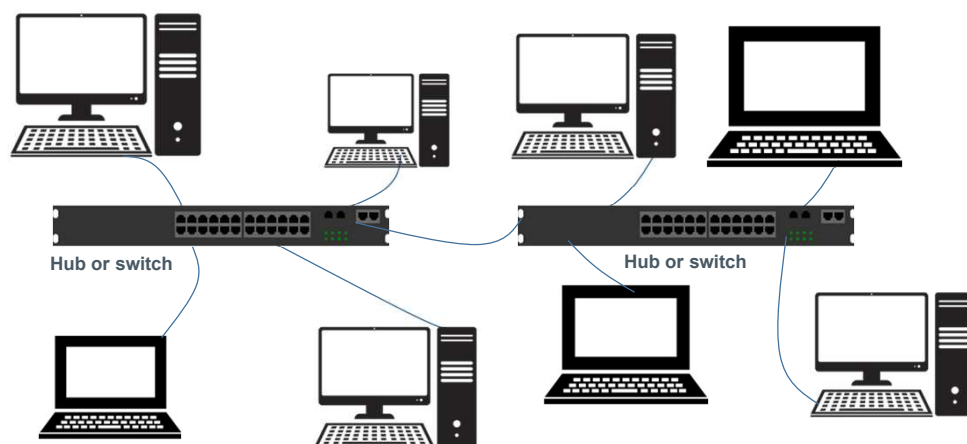
Router: device that connects two or more **networks**

- Connect the computer to the Internet
- Wireless router: provide wireless network access to the devices



Hub vs. Switch

A hub or switch is a device which provides a central point for cables in a network



MiS Router

A router connects multiple computers/devices or other routers together and transmits data to the destination on a network

- Through a router, the networks can share access to a broadband Internet connection, such as through a cable or DSL modem

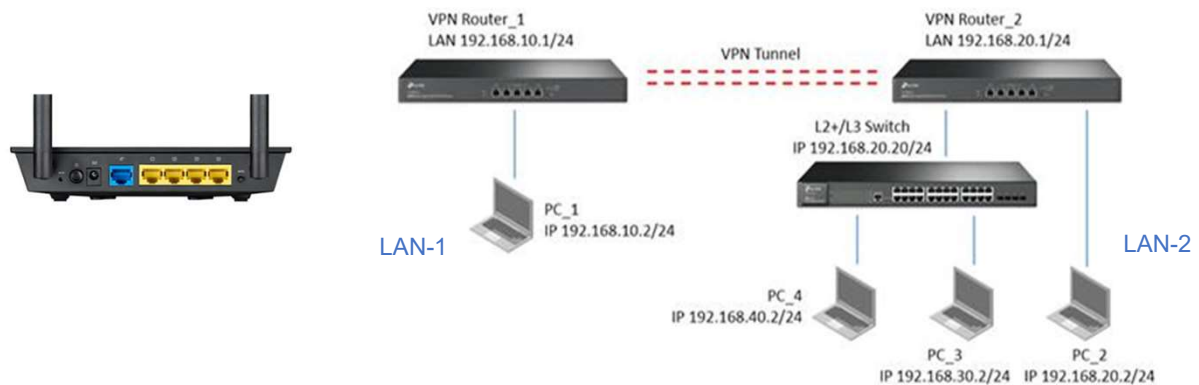


Image credit: <https://www.tp-link.com/tw/support/faq/2136/>

MiS Mac Address

Media Access Control (MAC) address identifies a unique network interface

MAC address is a 12-digit string

- Each digit can be any number (0~9) or a letter (b/t A and F)
 - **E1-A5-5E-24-34-EB**
- First six digits represent the **adapter's manufacturer**
- Last six digits represent the unique identification num for that **specific adapter**

The MAC address **contains no information about which network a device is connected to**

- Deny access to specific MAC addresses, can't do it with IP address



Mac Address & IP Address

IP address: where the target (like your home address)

- Assigned by ISPs and can be re-assigned as devices connect and disconnect
- Multiple devices (with virtual IPs) share a public IP
 - Many people live in the same place (same mail address)
- The IP address gets the data to your router

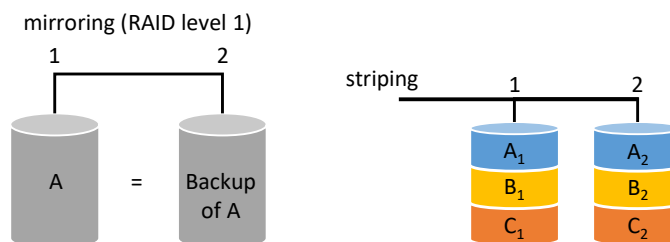
MAC address: who the target is (like your personal ID)

- Tied to a physical adapter and are assigned by manufacturers
- Every device on a router has a unique MAC address
 - Multiple personal IDs in your home
- MAC address identifies which device is which



Enterprise Storage - RAID

- Enterprise hardware allows large organizations to manage and store data and information with equipment designed for heavy use, maximum efficiency, and maximum availability
- **RAID** (redundant array of independent disks) is a group of two or more integrated hard drives
 - RAID duplicates data, instruction, and info to improve data reliability
 - RAID 0, 1, 5, and 10



Clip: <https://www.youtube.com/watch?v=U-OCdTzLac>



國立政治大學資訊管理學系

NCCU DEPARTMENT OF MANAGEMENT INFORMATION SYSTEMS

Introduction to Computer Science

Digital Security and Privacy Security

Shih-Yi (James) Chien

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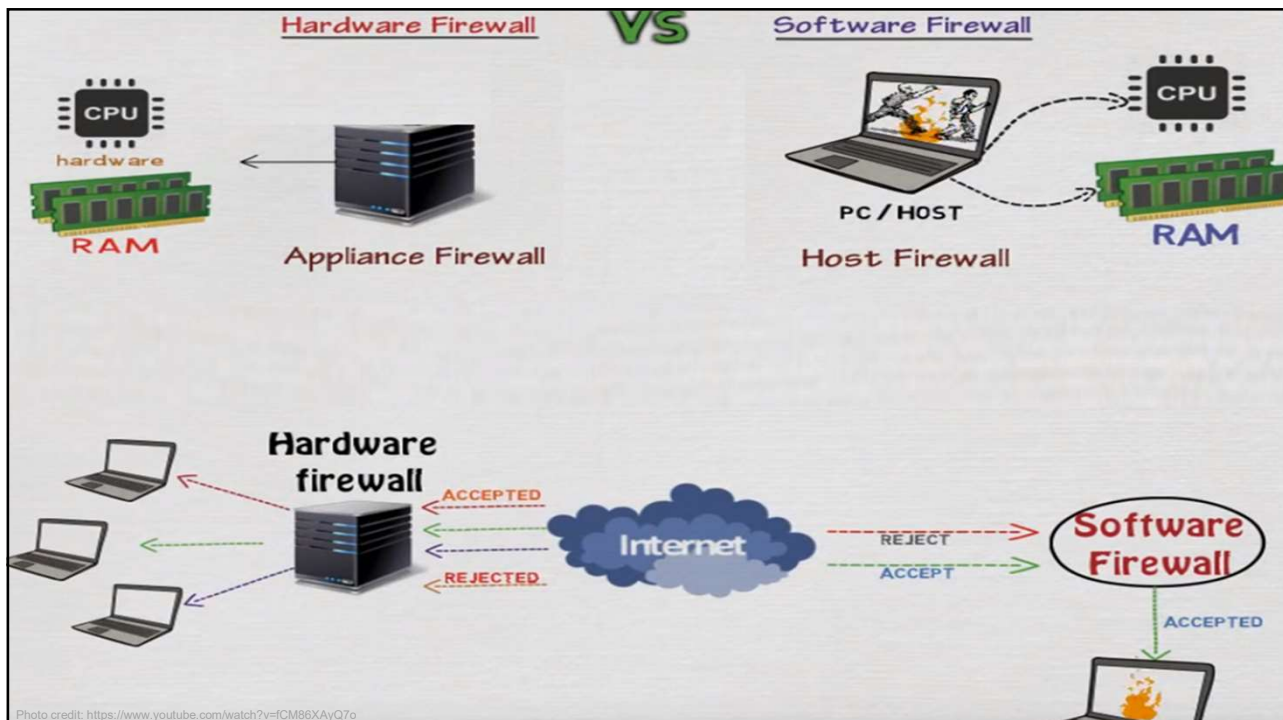
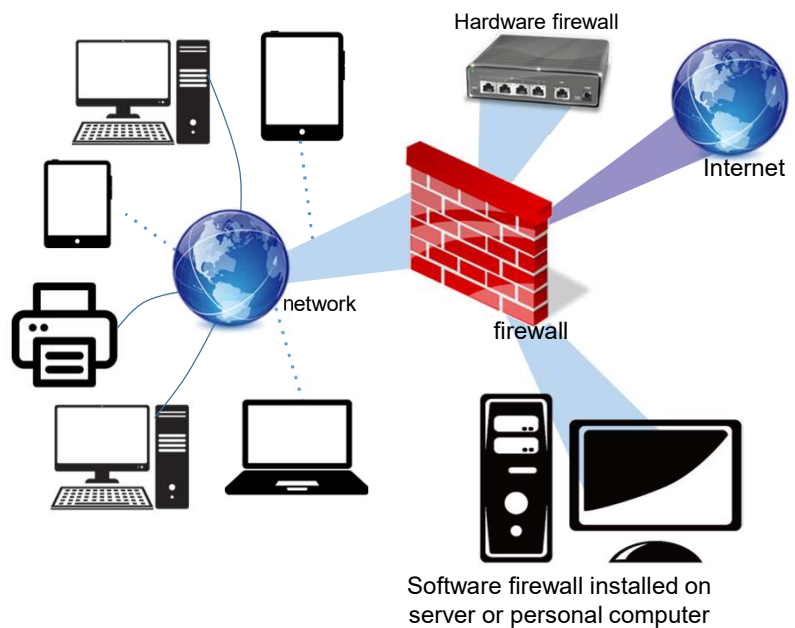
Key Points

1. What is a Firewall?
2. What are the differences b/t software and hardware firewall
3. What is encryption?
4. What are the differences b/t Symmetric and Asymmetric encryption
5. What is Digital signature? Why do we need it?



Firewall

Firewall is hardware and/or software that protects a network's resources from intrusion



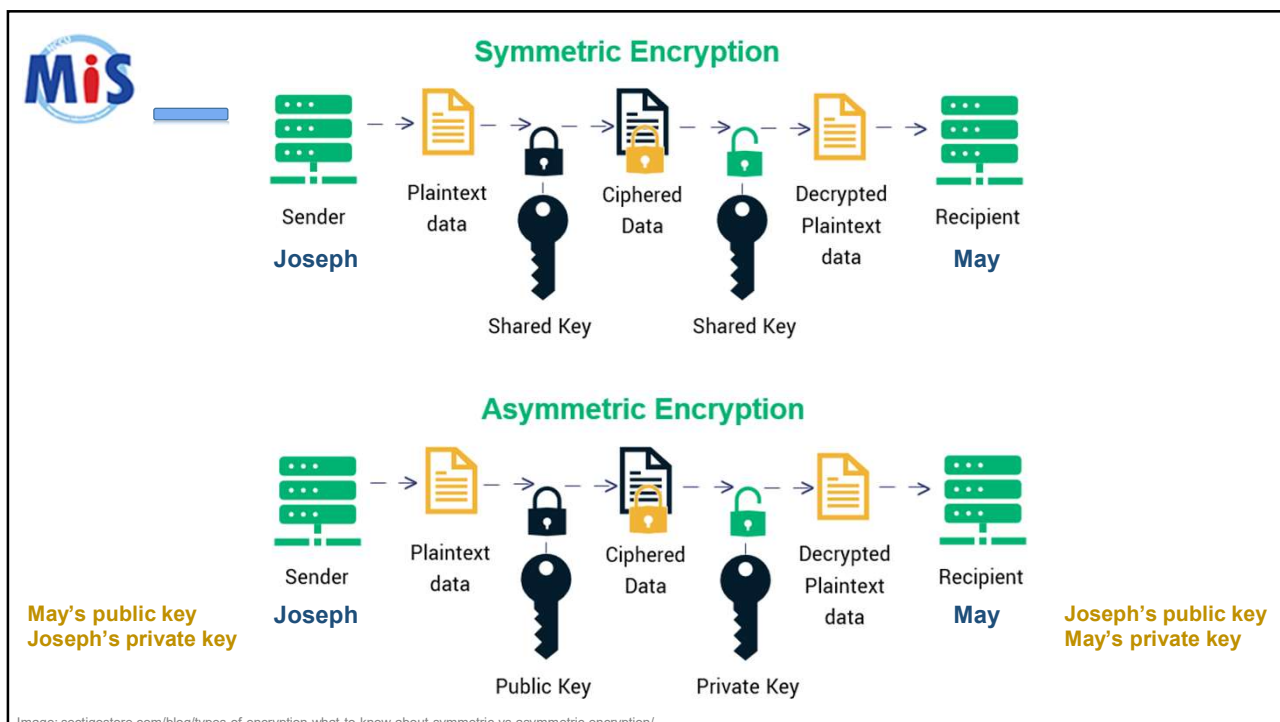


Encryption

Information theft occurs when someone steals personal or confidential information

To prevent information theft, **encryption** is the process of converting human-readable data into encoded characters to prevent unauthorized access

- To read the data, the recipient must **Decrypt**
 - Imitation Game message decoded
 - <http://www.youtube.com/watch?v=C25CwNIVjA&t=3m5s>



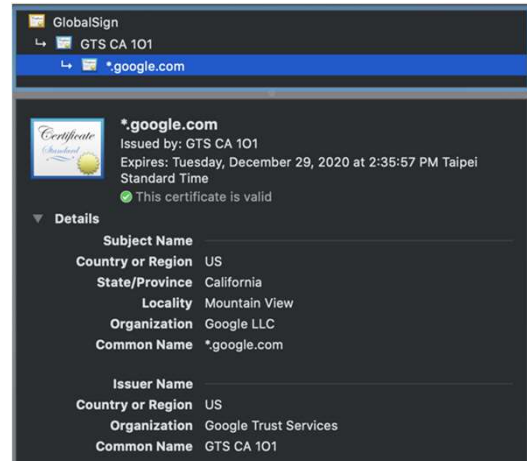


Digital Signature

Digital signature is an electronic, encrypted stamp of authentication on digital information

- Integrity: ensure the info originated from the signer and was not altered; any change made to the signed data invalidates the whole signature
- Nonrepudiation: enables you to prove to have received or sent a message from or to a trading partner

Digital certificate is a notice that **guarantees** a user or a website is legitimate



Pain is temporary; GPA is forever.

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